

DHANVA TEACH Clinical Research Report

Genetic Splice-Site Mutations in Familial Hypertrophic Cardiomyopathy

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Abstract: Genetic Profile of Familial Hypertrophic Cardiomyopathy

Hypertrophic Cardiomyopathy (HCM) is an autosomal dominant genetic cardiac disorder characterized by unexplained left ventricular hypertrophy in the absence of other cardiac or systemic diseases. This clinical review examines mutations in sarcomere protein genes, specifically focusing on the cardiac myosin-binding protein-C (MYBPC3) gene and its implications for clinical risk assessment.

Sarcomeric Gene Mutations & Pathophysiology

Mutations in MYBPC3 represent approximately 30% to 40% of all identified genetic variants in familial HCM. Most MYBPC3 mutations are frameshift, splice-site, or nonsense mutations that lead to a C-terminal truncated protein, causing haploinsufficiency. This disruption of the sarcomeric structure impairs myofilament contractility, triggers secondary hypertrophic signaling pathways, and promotes interstitial fibrosis.

Clinical Evaluation and Risk Stratification

Evaluation of suspected HCM patients requires a 12-lead ECG, transthoracic echocardiogram (TTE), and cardiac magnetic resonance (CMR) imaging to quantify maximum wall thickness and identify late gadolinium enhancement (LGE). Risk stratification for Sudden Cardiac Death (SCD) must assess prior cardiac arrest, family history of premature SCD, unexplained syncope, non-sustained ventricular tachycardia (NSVT), and massive left ventricular hypertrophy (thickness $\geq 30\text{mm}$).

Therapeutic Interventions & Management Guidelines

First-line pharmacotherapy for symptomatic obstructive HCM consists of beta-blockers (e.g., metoprolol) or non-dihydropyridine calcium channel blockers (e.g., verapamil) to reduce the heart rate and prolong diastolic filling. For patients with drug-refractory severe symptoms and a left ventricular outflow tract (LVOT) gradient $\geq 50\text{ mmHg}$, surgical septal myectomy or alcohol septal ablation is indicated. High-risk patients should receive an Implantable Cardioverter-Defibrillator (ICD) for primary prevention.